

Professional Summary

Seasoned Linux System Engineer with experience in fast paced Unix/Linux environments and expertise in several flavors of Linux including Red Hat, CentOS and Ubuntu. Excellent installation, configuration, maintenance and system's users administration skills. Track record of accomplishing extraordinary results in working technicians from various engineering disciplines to troubleshoot complex system-level issues and working with organizational and technical Linux.

Strong knowledge on the MVC architecture of Angular 6 in developing apps, controllers, and directives, customized to meet the needs of the application. Delivered native functionality for the applications using Flutter framework such as storage, camera, and splash screens. Experience in using Mobile Frameworks like Native Android (Java & Kotlin) and Flutter for iOS, Android, and WEB Platforms.

Experience in Microsoft Hyper-V, experience in managing and optimizing virtualized environments. My expertise includes configuring and maintaining Hyper-V virtual machines, implementing high availability and failover clustering for minimal downtime, and utilizing Hyper-V Replica for robust disaster recovery solutions. I am proficient in automating tasks and managing virtual infrastructure using PowerShell, ensuring efficient operations and resource management. Additionally, my skills extend to configuring virtual networking, managing storage solutions, and applying security best practices to safeguard virtual environments. My hands-on experience with Hyper-V enables me to deliver reliable and scalable virtual server solutions tailored to organizational needs.

Full Stack Developer with a background in Linux system engineering and DevOps, I bring a unique blend of skills that bridge the gap between development and operations. My experience in Linux system administration equips me with a strong understanding of server management, performance optimization, and security best practices, which I leverage to create robust web applications. Proficient in front-end technologies such as HTML, CSS, and JavaScript frameworks (React, Angular), I develop user-friendly interfaces that enhance user experience. On the back end, I utilize programming languages like Python and Node.js, along with database management systems like PostgreSQL, to build scalable and efficient applications.

DevOps expertise allows me to implement CI/CD pipelines using tools like Jenkins and GitLab, ensuring smooth deployment processes and minimizing downtime. Additionally, I have hands-on experience with containerization using Docker and cloud services like AWS, which enhances application scalability and reliability. My collaborative approach, combined with my problem-solving skills, enables me to work effectively in Agile environments, ensuring that development aligns with operational needs and user requirements.

Education

Bachelor Of Science (B.S: Informations Systems Management - University Of Maryland, College Park, MD

Experience

Devops Engineer
University of Rochester

November 2022 -
Current

- Experience in Design, building, documenting, implementing, and operationalizing highly scalable, available, reliable APIs that are continuously deployed and updated through an automated CI/CD pipeline. Enable user analytics, logging, monitoring, and instrumentation for all services
- Installed, Configured CI/CD for the Application development process
- Experience with Linux servers in virtualized environments

- Experienced with REST API, SQL Server, AWS, iOS & Android, Flutter & Dart
- Experienced in App Architecture patterns like MVVM, MVP & MVC
- Deep understanding of the Dependency Inversion principle and its implementation with Flutter
- Experienced in developing application code using Java / JS / Python and building Android from AOSP
- RedHat Enterprise Linux installation of virtual and physical servers (RHEL 6, 7, 8)
- Install and configure Apache on RedHat/UNIX servers
- Performing Unix configuration changes and software installations
- Involved in loading data from the UNIX file system to HDFS.
- Linux (RedHat 6/7/8) UNIX System, IBM AIX, Server
- Solid knowledge of architectural patterns addressing better code reuse, maintainability, and ability to implement with Flutter
- Managed a team of highly motivated developers and develop flagship IoT and Customer focused mobile apps
- Installations, configurations, upgrades and migrations on Linux and AIX.
- Supported CIS and Engineering organizations with various flavors of Unix and Linux
- Installed, configured and migrated all Unix servers
- Secured and patched UNIX servers
- Developed user-friendly web interfaces for internal tools using HTML, CSS, and JavaScript (React).
- Built Python-based automation scripts transitioned into web applications using Flask, improving system management efficiency.
- Managed PostgreSQL databases to optimize performance metrics for reporting applications.
- Created RESTful APIs using Node.js and Express.js for seamless data integration between internal systems.
- Utilized Git for version control and collaborated effectively in Agile teams.
- Implemented Docker for containerization, ensuring consistent application deployment across environments.
- Set up CI/CD pipelines using Jenkins to streamline deployment processes on AWS.
- Developed High Availability and Disaster Recovery for the UNIX environment.
- Strong Scripting skills in Bash, PowerShell, Perl, YAML, Json, Ruby and Python scripts to automate the deployments and for environmental changes.
- Developing scripts for build, deployment, maintenance and related tasks using Jenkins, Docker, Maven, Python and Bash.
- Responsible for continuous integration (CI) and continuous delivery (CD) process implementation - using Jenkins along with python and shell scripts to automate routine jobs.
- Experienced in implementing Splunk, Puppet, C DevOp.
- Experience configuring and managing Puppet master server and also experience in updating and creating modules and pushing them to puppet clients.
- Implemented automation tools such as Splunk, Puppet, Chef and DevOps
- Created Python scripts to automate AWS services which include web servers, ELB, Cloudfront distribution, databases, security groups and application configuration.
- Involved in maintaining and editing Python scripts for application deployment automation.
- Wrote serverless applications (Lambda functions) in python, node.js to perform various transformations and analytics on large data sets in EMR clusters.
- Configured NFS resources on RHEL 6,7,8 and UNIX
- Hands-on experience on Azure VPN-Point to Site, Virtual Networks, Azure Custom security, Endpoint Security, firewall, Windows Azure name resolution, Scheduler, Automation and Traffic Manager.
- Work experience in Azure App & Cloud Services, Paas, Azure Data Factory, Azure SQL Data Warehouse, Power BI, Azure Blob Storage, Web API, VM creation, ARM Templates, PowerShell scripts, Iaas, Lift & Shift, Storage and database.
- Managed servers on the Microsoft Azure Platform
- Azure Virtual Machines instances using Puppet
- Configuration Management. Created Puppet manifests, profiles and roles modules to automate system operations.
- Creating and maintaining containerized micro services and configuring/maintaining private container registry on Microsoft Azure for Hosting Images and using
- Created and used Dart and Flutter packages and plug - ins, Flutter widgets, and State management (Provider, BLoC, etc.)
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- Extensively worked with making calls to REST APIs to fetch data from the backend as JSON response and parse it on the UI for DOM/CSS Manipulation.
- Deep understanding of tools and techniques related to code analysers, layout inspector, performance view, memory view, multithreading, networking & Security, etc
- Worked with XML-related technologies such as XML, XSL, DOM, JDOM, SAX, XSD, XSLT, and JAXB and used XML Schema for retrieving patient details in some of my projects.

- Used Hibernate and JDBC to connect to databases like Oracle, MYSQL, and DB2 to store, delete, manipulate, and retrieve data from them in many of my applications.
- In-depth knowledge of integration of RESTFUL Web Services, HTTP Services, and Web Services with web applications
- Azure Services: Azure Active Directory, Azure Functions, Azure Kubernetes Service (AKS), Azure Storage
- Scripting: PowerShell, Bash
- Cloud Modernization: Migrating legacy applications to Azure, containerization using Docker
- DevOps Tools: Azure DevOps, Jenkins, Terraform, Git
- Monitoring and Maintenance: Azure Monitor, Application Insights
- Led modernization efforts for on-premises applications, successfully migrating to Azure, resulting in a 30% increase in operational efficiency.
- Designed and implemented Azure cloud solutions, utilizing services like Azure Functions and AKS, to support scalable and resilient application deployments.
- Automated infrastructure provisioning and management using PowerShell scripts, reducing deployment times by 40%.
- Collaborated with cross-functional teams to establish DevOps practices, integrating CI/CD pipelines using Azure DevOps and Jenkins, enhancing release cycles.
- Developed and maintained monitoring solutions using Azure Monitor and Application Insights, improving system reliability and performance tracking.
- Experienced with version control and group repositories, comfortable working intimately on a codebase with other developers

- Windows Active Directory
- Reconfigured all ESX network connectivity utilization
- VLAN tagging and teaming.
- Experience in supporting Windows Cluster server running 2-4 nodes on Windows 2003/2008 & 2012 supporting applications like web servers, SharePoint and Active Directory environment
- Build VMware vSphere hosts and added to new/existing VMware Clusters on Intel Stack and involved in the Data Centre Transformation.
- Migrated the VMs to a Linux KVM / libvirt environment utilizing a CEPH storage cluster Wrote Automation scripts automate the provisioning of Azure resources like Virtual Machines, Virtual Networks, Traffic Manager, Storage, Service Bus and Scheduler make calls to Puppet scripts that provisioning / configuration of servers on the instances.
- Experience in using Docker and setting Azure Container Registry with Docker and Docker-compose and actively involved in deployments on Docker using Kubernetes.
- Performs hands on configuration and management of
- AWS services via the dashboard and CLI console. Work with development, security and operations teams to provision AWS resources as needed to support application requirements.
- Develops and supports tools to facilitate automated management of AWS services and integration with existing management tools where appropriate.
- Installed and introduced onto the network Centos 6.10, Centos 7.6, RedHat 6.10, RedHat 7.6, RedHat
- Maintained user account management, configured servers to register with Redhat Network (RHN), monitored servers - workload assessment, service console monitoring
- Experience in monthly Patching event for Production and Non-Prod Linux servers through Redhat satellite, monthly patches channels
- Installed and configured several Redhat Enterprise Linux in both Physical Blade utilizing Kickstart and VMWare EXSi 5.0 virtual environment
- Converted our staging and Production environment from a handful of AMI's to a single bare metal host running Docker.
- Used Docker to virtualize deployment containers and push the code to EC2 cloud using PCF. Built additional Docker Slave nodes for Jenkins using custom built Docker images and instances.
- DevOps Practice for Micro Services using Kubernetes as Orchestrator.
- Kubernetes is used to orchestrate the deployment, scaling, management of Docker Containers.
- Automated the deployment of applications inside software containers and Configured Docker containers for branching purposes.
- Administer and migrate data on RHEL5,6,7 & 8 an servers
- Manage file systems like LVM, LVM extension & reduction on RHEL
- Provide support that can resolve incidents tickets and change request with RHEL5/6/7/8 related issues
- Install and configure RHEL on HP DL385-DL585 servers
- Experienced with Linux Variants; RHEL, CentOS, Ubuntu.
- Redhat Package management using Yum and RPM.
- Created Cron jobs as needed and Systems backup for RedHat Enterprise Linux, Centos based environments
- Worked across both private (OpenStack) and public Clouds (Amazon AWS).
- Debug and resolve Pre-Post OpenStack deployment failures.
- Installation and configuration of OpenStack Keystone, Glance, Nova, Neutron, Horizon, Cinder, Heat
- Installation and configuration of OpenStack on Centos
- Experience in networking protocol TCP/IP, LAN, FTP, TFTP, NFS, DHCP, SMTP, POP, IMAP, NTP, HTTP SO on.
- Remote administration and system configuration through scripting, Linux CLI
- Installed and responsible for GIT software version control for client environments. configuration of RedHat to fit security compliance using STIG, CRISP, SCAP
- Responsible for installing, configuring, and maintaining RedHat LINUX/UNIX operating systems
- Linux systems administration skills with emphasis on RedHat Enterprise Linux or Oracle Enterprise Linux
- Was working on integrating Puppet with a new RF satellite server.
- Installing RedHat Linux using kickstart and applying security policies for hardening the server based on the company policies.
- Creating Hardware Raid and LVM and administering them in RedHat Linux.
- Configuring RedHat systems over the network, implementing automated tasks through cron jobs, resolving tickets according to the priority basis.
- Installation of patches and packages using RPM and yum in RedHat Linux.
- Setup, configured, and debugged network
- configurations for RedHat servers and workstations
- Installed RedHat Linux using kickstart
- Provided 24-hour on-call support when scheduled.

- Served as a technical escalation point to others serving in on-call roles.
- Maintained a broad knowledge of technology, equipment, and/or systems
- Perform miscellaneous job- related duties as assigned
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- Developed system engineering, software engineering, system integration and distributed system architectures
- Improved cloud systems functionality with automation and expert configuration management.
- Monitored applications and infrastructure with observability technologies and managed alerts.
- Configured and maintained CI/CD pipelines with Jenkins and GitLab.
- Collaborated with data processing personnel and project managers to gather data processing limitations or capabilities
- Determined feasibility of proposed designs within time and cost constraints.
- Developed and directed software system testing and validation procedures, programming and documentation
- Designed, deployed and maintained systems running Red Hat Linux and CentOS.
- Hyper-V Virtual Machine Management
- Configured and managed Hyper-V virtual machines, including resource allocation, disk provisioning, and virtual network setups.
- Installed and optimized various guest operating systems (Windows, Linux) within Hyper-V environments.
- High Availability and Failover Clustering:
 - Set up and maintained Hyper-V failover clusters to ensure high availability of virtual machines and minimize downtime.
 - Performed live migration of VMs between Hyper-V hosts without service interruptions.
- Hyper-V Replica and Disaster Recovery:
 - Configured Hyper-V Replica for disaster recovery purposes, ensuring replication of critical virtual machines between hosts for failover readiness.
- PowerShell Automation in Hyper-V:
 - Automated VM deployment and configuration using PowerShell scripting within Hyper-V environments to streamline operations.
 - Developed PowerShell scripts to monitor VM health, manage resource allocation, and automate routine tasks.
- Virtual Networking in Hyper-V:
 - Configured virtual switches, VLANs, and network adapters in Hyper-V to create isolated and shared networks between virtual machines.
 - Managed Hyper-V networking setups to ensure secure and efficient communication between virtual machines and the physical network.
- Storage Management with Hyper-V:
 - Created and managed virtual hard disks (VHD/VHDX) and configured storage spaces in Hyper-V environments.
 - Implemented shared storage solutions using iSCSI and SMB for Hyper-V clusters.
- Security and Hardening in Hyper-V:
 - Applied Hyper-V security best practices, including configuring Shielded VMs, Secure Boot, and managing access control for VMs.
 - Hardened Hyper-V hosts and virtual machines by implementing security baselines and monitoring for vulnerabilities.

Linux Administrator
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March 2021 - September
2022

- Install and configure Redhat/CentOS linux on virtual environment
- Experience in installation, configuration, support of Linux (Redhat) and Windows operating systems
- Provide technical support for both hardware and software issues for end-users
- Create files and directories at user and root level
- Manage multiple operating systems and technologies like Linux, Windows and SQL database

- Alert management from the enterprise monitoring tools, escalations from IT Service Desk
- Extensive experience of addressing Linux systems operations center incidents, reporting, analyzing and resolving production issues
- Perform incident analysis like process management, CPU and memory analysis to quickly recover from service interruptions, and to prevent recurring of issue
- Experience with logical volume management (LVM), analyzing the disk usage and create disk partitions
- Respond to high volume of support tickets reported by service desks
- Experience in file system maintenance like responding to file system alerts, finding the file system usage and analyzing the highest offender and increasing file systems space
- Managing, installing, and, upgrading software packages using YUM and RPM and creating YUM repository files for offline servers
- Perform and manage system backups and replications running on 3rd party backup solutions and through local RHEL tools (e.g Tarball, gzip, dd etc)
- Work with Service Desk and other Support teams to quickly resolve high impacting incidents
- Monitored systems, networks and infrastructures using tools such as SiteScope, Zabbix or Nagios
- Support other RHEL versions (5, 6, 7 and 8) running in production and development environments
- Setup and administer user and groups accounts along with setting permissions for user
- Limit user account privileges by configuring sudoers
- Performed kernel tuning based on the applications to be installed and the number of users
- Schedule various regular, periodic, future and queue tasks by using crontab
- Work on shell scripts for CPU usage, Memory consumption and number of I/O to collect performance stats and troubleshoot issues on RHEL 5/6/7/8
- Install and configure SSH key-based authentication
- Managed break/fixes through ticketing solutions
- Deploy and manage Linux VMs on ESXi server using vSphere client
- Install, administer and troubleshoot NFS, DNS, and FTP servers
- Provide ongoing support on networking services and protocols such as TCP/IP, NIS, DNS, NFS, LDAP,
- FTP, SSH, SNMP, SSL, and HTTP
- Front-End Development
- HTML/CSS: Developed and maintained web interfaces for internal tools used by the IT department, ensuring a user-friendly layout and adherence to accessibility standards.
- JavaScript: Implemented dynamic features on the company's intranet using JavaScript frameworks (like React) to enhance user experience and streamline workflows.
- Responsive Design: Created mobile-responsive dashboards for monitoring server performance, utilizing frameworks like Bootstrap to ensure compatibility across devices.
- Back-End Development
- Programming Languages: Leveraged Python to build automation scripts for system administration tasks, transitioning them into full-fledged web applications using Flask to provide a user interface for colleagues.
- Frameworks: Used Node.js and Express.js to develop RESTful APIs for integrating system monitoring tools, allowing for real-time data visualization on the front end.
- Database Management: Managed PostgreSQL databases to store logs and performance metrics, optimizing queries to improve data retrieval times for reporting applications.
- API Development: Created APIs for communication between different internal systems, facilitating seamless data flow and improving operational efficiency.
- Version Control Systems: Utilized Git for version control in collaborative projects, maintaining clear commit histories and facilitating code reviews with team members.
- Containerization: Implemented Docker to containerize applications, ensuring consistent deployment across different environments and simplifying the setup process for development and testing.
- CI/CD Pipelines: Set up Jenkins to automate the deployment of applications, reducing deployment time and minimizing human error during the release process.
- Cloud Services: Deployed applications on AWS, utilizing services like EC2 for hosting and RDS for database management, ensuring high availability and scalability.
- Implementation and Integration of NTP for time synchronization (ntp and chronyd)

- Perform server refresh and maintain server integrity by configuring firewall rules for network communication
- Configure dynamic and static network settings for IPv4 and filter packets
- Install, configure and manage Apache (HTTPD) on Linux production servers
- Perform error analysis and recovery, documentation and procedures required for reference
- Provide 24x7 on call duties on a rotation basis with the other system engineers as part of incident management
- Participate in Change Control Management and documentation of the procedures
- Experience in decommissioning of servers and monitoring inventory
- Install and configured RAID 0/1/5 at hardware level
- Coordinate with Application support, Server Operations, Network Operations, Middleware and Database
- Support teams in troubleshooting production issues
- Maintain systems by keeping them compliant with company standards for optimum performance and capacity utilization
- User files permission management including assigning, adding and updating permissions at the file and directory level
- Provision UNIX accounts, groups, and assignment of accounts to groups
- Initiate and facilitate technical bridges as per businesses need for command center
- Everyday use of filters and text processing commands (cut, sort, grep, awk, uniq, wc)
- Keep the system secure by monitoring user activity through secure log messages and tools such as who, last, etc
- System management using commands (top, df, dmesg, iostat 1, netstat)
- Setup aliases for ease of management
- Manage OS network (ping, ifconfig, netstat, tcpdump, networking config files)
- Troubleshoot issues with VMs running on VMware ESXi 5.5 and 6.0
- Install, configure and maintain Windows OS versions 2012 R2 (or later), including OS patching, drivers and hardware firmware
- Demonstrate engineering excellence in the design, systems management, automation, patching, deployment, troubleshooting, and performance tuning for Linux based operating systems
- Write and maintain documentation such as instruction guides, troubleshooting procedures, incident process, SOW, Root cause analysis etc
- Created and manage Solaris Jumpstart and Linux Kickstart servers and processes to automate and standardize the installation process, reducing installation time by 35% and post-installation errors by 50%
- Document and implement a disaster recovery plan that included backup schedule policies, regular testing, and off-site recovery plans to ensure data integrity and assurance
- Resolve security vulnerabilities by analyzing and recommending improvements in communications and network security at the component level
- Eliminated extended downtime of network and processes by installing and configuring a Nagios server to monitor
- Windows/Linux servers

Linux Technical Support

April 2020 - April

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2021

- Assisted customers having hardware, software and networking issues
- Assisted business customers with LAN /WAN / Static Ip configuration issues
- Troubleshoot customers issues with home health check / node health check tools for possible outages
- Troubleshoot any problems or issues that arise with Windows desktops/laptops from the network drop to the desktop/laptop prioritizing systems infected with viruses
- Maintained updated knowledge of company products and services to better provide customer support and service solutions
- Gathered customer and technology information to determine technical support level; escalated issues to the appropriate department as necessary
- Configured and deploy desktops/laptops for new hires and IMAC service for office moves
- Created spreadsheet reports regularly as required by manager for asset management, asset information of new hires' devices, etc

- Assisted regional IT teams with Smart Hands support in the facilities such as IDF/MDF closets, Data centers and other related server rooms
- Provide deskside support in person, on the phone and remote using LogMeIn, Bomgar and Skype
- Utilized Active Directory to add/remove users, unlock accounts and reset passwords
- Resolve tickets and request in in timely manner
- Sustain network integrity, server deployment and security
- Actively participate in process improvement initiatives
- Work cross functionally to develop, implement and support new end user solutions
- Assist in migration existing services from physical data centers to AWS cloud
- Analyzed equipment performance records to determine need for repair and replacement
- Configured email applications or virus protection software
- Operated master console to monitor performance of computer systems and networks and to coordinate computer network access and use

Projects

SelloSoft *Sep 2023 – Present*

SelloSoft is a **SaaS-based Point of Sale solution** tailored for diverse industries, including food and beverage, retail, beauty and health, hospitality, and services. Our innovative software seamlessly integrates with cutting-edge hardware devices provided by **SUNMI**, a leading Chinese technology company.

At SelloSoft, I have been involved in the project from the first lines of code. My responsibilities include developing the mobile app for SelloSoft, sprint planning, API integration, and writing reusable code.

MenderMended *Oct 2023 – Dec 2023*

MenderMended, a **mental health platform** for psychiatrists and patients. Mender streamlines appointment management for psychiatrists, while mended allows patients to buy tokens, schedule appointments, and join support groups. This app merges mental health and technology, offering a

User-friendly solution for professionals and individuals seeking support. My role was to develop the responsive frontend for the web version of the MenderMended mobile application.

FreshOven *Dec 2022 – July 2023*

Recipe Sharing and Home Kitchen Flutter based Mobile Application. FreshOven ecosystem consists of 3 Mobile Apps. It was my final year project for my bachelor's degree.

Technologies Used: Flutter, Dart, NodeJS, Express, JavaScript, MySQL, Postman, JSON, Android Studio

Skills

- Coordinated with vendors and company personnel to facilitate purchases
- Computer networks and related
- Customer needs to identify, Requirements
- Performed routine network
- Control records
- Configurations and determine To such changes
- Problems
- Whether adjustments needed to Made in future.
- Windows, VMWare, Oracle
- Scripting in bash
- Management, FTP, NFS, Etc.
- MySQL etc.
- Excel,
- Etc
- IBM AIX
- Optimization
- Application Configuration
- Red Hat Enterprise Linux
- Software System Design
- Final Design Delivery
- Multi-Factor Authentication
- Puppet
- CyberArk
- MS HyperV
- Loaded data and installed software and peripherals
- Computing environments
- Predict, interpret and evaluate
- Performed data backups and
- Startup and shutdown
- Recommended changes to
- Hardware and
- Conferred with network users to
- Monitored network
- Be made and to determine
- Operating System:
- Virtual Box
- Networking: Static IP
- NTP/Chronyd,
- Database: Exposure to Microsoft
- OS Applications:
- PowerPoint, Linux Sendmail,
- Maintenance Organization and
- Design Solutions
- Network Security
- Design Improvements
- Configuration and Management
- Cloud Hosting
- Script Writing
- IP Addressing and Subnetting
- Nagios
- Maintained and administered
- Gathered data pertaining to
- System and network
- Disaster recovery operations
- Procedures and maintained
- Improve systems and network
- Software requirements related
- Resolve existing system
- Performance to determine
- Where changes needed to be
- CentOS/Redhat,
- Computer Languages: Linux
- Assignment, NIC
- HTTPD, DNS/Bind, rsyslog etc
- Access or
- Microsoft Word,
- DNS, SSH, SCP,
- Development
- Performance Monitoring and
- Kernel Patches
- Wide Area Network (WAN)
- Security Needs Assessment
- Computer Operating Systems
- Linux Operating System
- Testing and Quality Assurance
- Vsphere